RV College of Engineering®

Department of Electronics & Communication Engineering

Date: 19/03/2024	**Test - 3**	**Max. Marks:50**
Semester: III	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course: Analysis and Design of Digital Circuit with HDL		**Code: EC234AI**

S No	Questions	M	BT	CO
1. a)	Design a 4-bit universal shift register for the following conditions.	6	3	3
b)	Design MOD-5 asynchronous counter.(Assume CLEAR input is available in the IC)	4	2	1
2.	Design a ripple decade counter using JK-FFs without using PRESET and CLEAR signals with the help of a state diagram and explain. In the same design, show that how MOD-2 and MOD-5 counters can be built to support the design of IC-7490.	10	4	3
3.	Design an arithmetic and logic unit with three selection variables $S_2S_1S_0$ that generate the following operations. Draw the logic diagram of one typical stage.	10	4	4
4. a)	Construct a 4-bit Ring counter using D-FF's. Draw the timing diagram for the same. Write the Verilog code for the same.	5	3	2
b)	What is Scratch pad memory? Analyse the execution sequence in a processor unit with 2-port memory. Draw relevant block diagram.	5	2	2
5.	For the given timing diagram as shown in Fig5 state the sequence which is detected and Develop a Verilog code for mealy FSM, assuming that overlapping of pattern is allowed.	10	3	2

S_0	S_1	Operation
0	0	No change
0	1	Shift Right
1	0	Shift Left
1	1	Parallel load

Arithmetic Operations					Logical Operations			
S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$
0	0	0	F=A	F=A+1	1	0	0	OR
0	0	1	F=A+B	F=A+B +1	1	0	1	XOR
0	1	0	F=A-B-1	F=A-B	1	1	0	AND
0	1	1	F=A-1	F=A	1	1	1	Complement

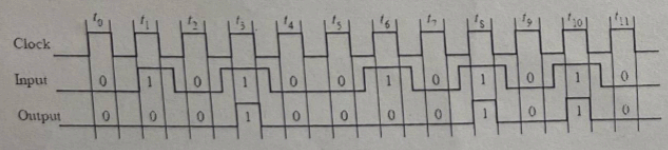

Fig5